

Analyzing Query Embedding Spaces in RAG-Based Text-to-SQL Systems Using PCA and SVM

Bridging Mathematics and AI

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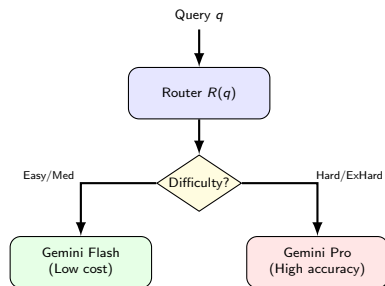
The Core Problem: Enterprise Query Routing

The Challenge in Text-to-SQL RAG:

- **Text-to-SQL:** Translating natural language queries $q \in Q$ into executable SQL code.
- **Resource Trade-off:** Large Language Models (LLMs) are accurate but expensive and slow. Small models (SLMs) are fast and cheap but fail on complex schema joins.

Mathematical Goal: Find a routing function $R : Q \rightarrow \{M_{\text{small}}, M_{\text{large}}\}$ that minimizes inference cost and latency $C(R(q))$ subject to a target accuracy threshold α :

$$\min_R \mathbb{E}[C(R(q))] \quad \text{subject to} \quad \text{Acc}(R(q)) \geq \alpha$$



Text Vectorization

- We use a pre-trained **Sentence Transformer** model currently running on Sumvec's live platform.
- The model maps a natural language query to a dense vector:

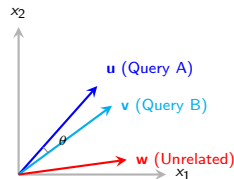
$$f : \text{Query} \rightarrow \mathbf{v} \in \mathbb{R}^{768}$$

- Each of the 768 numbers is a coordinate representing a latent semantic feature.

Cosine Similarity To measure semantic similarity, we compute the angle θ between vectors using the inner product:

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$$

Semantic Proximity in $\mathbb{R}^2$



Similarity Retrieval

- Longer sentences naturally produce vectors with larger magnitudes.
- To compare queries purely by meaning, we must eliminate the effect of length and isolate vector **direction**.

Vector Norms For a vector $\mathbf{x} \in \mathbb{R}^{768}$, its Euclidean (L_2) norm is:

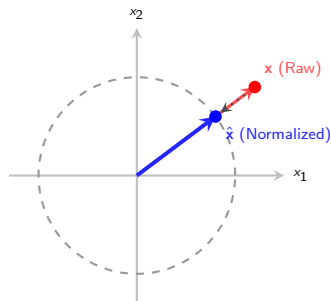
$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{768} x_i^2}$$

We normalize the vector to unit length (1.0):

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \implies \|\hat{\mathbf{x}}\|_2 = 1$$

This simplifies cosine similarity to a simple dot product: $\hat{\mathbf{u}} \cdot \hat{\mathbf{v}}$.

Projection onto the Unit Circle



Dataset Comparison:

Metric	Spider	Pinecone
Dataset Type	Benchmark	Production
Observations	10,000	11,435
Embedding Dim	384D	768D
Complexity	Static	Live Traffic

Heuristic Labeling (S): Based on SQL keyword counts:

- JOINS, GROUP BYs, Subqueries
- S score mapping to labels

Complexity Distribution:

Difficulty	Criteria	Empirical %
Easy	$S = 0$	$\approx 25\%$
Medium	$S = 1$	$\approx 33\%$
Hard	$S \in \{2, 3\}$	$\approx 27\%$
Extra Hard	$S \geq 4$	$\approx 15\%$

Statistical Imbalance: In both academic benchmark and live production traffic, the **Extra Hard** class represents only $\approx 15\%$ of the total data.

- Models naturally over-predict the majority classes.
- Requires balancing class weights in optimization.

The Curse of Dimensionality & Redundancy

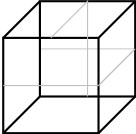
Curse of Dimensionality:

- **Volume Expansion:** The volume of a hypercube scales exponentially (s^d).
- **Data Sparsity:** In high dimensions ($d = 768$), data points become extremely isolated.
- **Distance Convergence:** Distances between random points converge to a constant value, making neighborhoods unstable.

Spatial Sparsity (2^d)

 1D: $2^1 = 2$ units

 2D: $2^2 = 4$ units

 3D: $2^3 = 8$ units

Semantic Noise & Redundancy:

- A 10% loss in variance represents "noise"—variations in punctuation or phrasing that do not change query intent.
- Real information lies on a much lower-dimensional manifold.

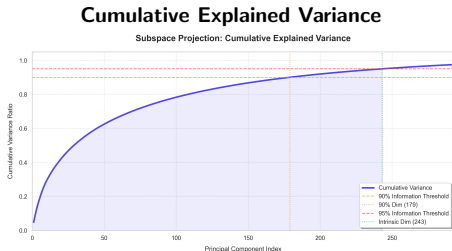
Principal Component Analysis (PCA)

PCA as a Change of Basis: A vector can be represented using different bases.

- PCA finds the **best orthonormal basis** for the data.
- The first basis vector points in the direction of maximum variance (spread).
- Each subsequent basis vector is orthogonal to the previous ones.

Production Insights (768D):

- Retaining **90% variance** requires only **220 components**.
- Represents a **71% dimensionality reduction** in production.



Benchmarking 4 PCA Variants (384D Spider Dataset): We run a timer to evaluate the speed, RAM efficiency, and accuracy trade-offs:

PCA Variant	RAM / Complexity	Trade-off & Feature
Standard	Heavy on RAM	Loads all data; fastest and most accurate.
Incremental	Very low RAM	Processes in mini-batches; nearly identical results.
Sparse	Slowest	Forces weights to zero; highly interpretable.
Kernel	High memory complexity	Maps to higher dimensions for curved patterns.

Operational Verdict: For query routing, standard PCA remains the baseline choice because of its high execution speed, while Incremental PCA provides the necessary memory scalability for large streaming databases.

Subspace Projection & Geometric Blind Spots

Linear Subspace Projection (768D $\rightarrow$ 2D):

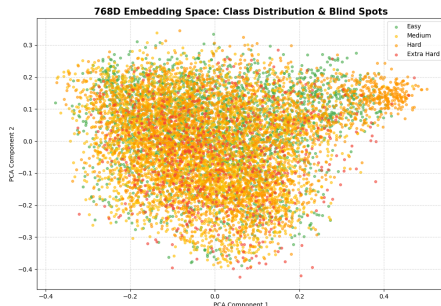
- We use a linear transformation to project the 768D space onto a 2D plane.
- **Analogy:** Shining a flashlight on a 768D object and observing its 2D shadow.

Key Geometric Insights:

- **Easy Queries:** Tightly clustered (small distance variance), meaning they share highly repetitive linguistic structures.
- **Extra Hard Queries:** Geometrically dispersed (outliers), representing "blind spots."

Production Resolution: Augmenting the database index with diverse nested-query examples to cover these blind spots.

768D Embedding Space Projection



Goal: Drawing Boundaries We train a Support Vector Machine (SVM) to draw decision boundaries between query difficulty levels.

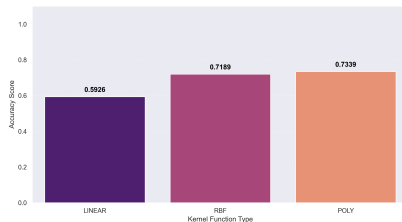
Kernel Comparison (50 Components):

- **Linear Kernel (59.3%):** Finds a straight boundary. Fails because class overlap is highly non-linear.
- **Polynomial Kernel (71.9%):** Slower to execute (5.8s) and inconsistent on live production traffic.
- **RBF Kernel (73.4%):** Chosen as our production standard. Fastest (3.8s) and most stable.

Why RBF Works: It uses a "local" approach that prioritizes nearby points, capturing tight "pockets" of Easy queries.

SVM Kernel Accuracies

Comparison of SVM Kernel Accuracies (50 Components)



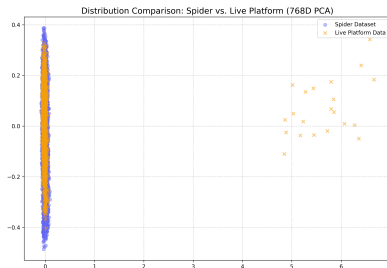
Performance Evaluation:

Metric	384D (Spider)	768D (Production)
SVM F1-Score	0.71	0.81
Routing Accuracy	79%	89%

Key Insights:

- **10-Point F1-Score Jump:** Moving from academic benchmarks to production vectors yields a massive boost.
- **Superior Separability:** Higher-dimensional production vectors ($\mathbb{R}^{768}$) spread out the semantic features, reducing linguistic overlap and allowing cleaner decision margins.

Separation: Spider vs. Live Platform



Operational Verdict:

- Standard PCA is the evidence-backed choice for Sumvec's database index.
- Eliminating the **71% redundancy** yields faster similarity search and lower cloud costs.

Query Routing Rule:

- **80% of Traffic:** Rated Easy/Medium, routed to free models via Openrouter.
- **20% of Traffic:** Rated Hard/Extra Hard, routed to Claude Haiku.

Production Cost Savings:

Feature	Raw	PCA	Savings
Dimensions	768	220	~ 71%
Data Size	37.2 MB	10.9 MB	26.3 MB

- Reduces vector search latency.
- Lowers memory footprint on Pinecone.
- Meets our optimization constraint: $\text{Acc}(R(q)) \geq \alpha$.